

Segment-sum concavity for projectively unimodular zonotopes

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Status. Substantial partial result; current verdict: likely valid. The source conjecture remains open for general zonoids.

1 Source problem

Fradelizi, Madiman, Meyer, and Zvavitch ask the following in Section 5 of [1]. If Z is a zonoid in $\mathbb{R}^n$,

$$L = \sum_{j=1}^m [0, u_j], \quad m \leq n, \quad P(t) = \text{Vol}_n(Z + tL),$$

is $P(t)^{1/m}$ concave for $t \geq 0$? Their exact passage appears below.

The main tool in the proof of Lemma 5.3 is Minkowski's second inequality, which relies on the fact that the polynomial $Q(t) = |K + tL|$ raised to the power $1/n$ is a concave function for $t \geq 0$. We conjecture that the concavity properties of this polynomial can be improved for Z being a zonoid and L being a finite sum of $m \leq n$ segments. More precisely, we conjecture that for any $m \leq n$ and any sequence of vectors $u_1, \dots, u_m$, $m \leq n$ from $\mathbb{R}^n$ if $P(t) = |Z + t \sum_{i=1}^m [0, u_i]|$, then $P^{1/m}(t)$ is a concave function for $t \geq 0$. This conjecture would follow if the statements of Theorem 5.2 or Theorem 3.9 would be true. Still we can prove the following proposition in $\mathbb{R}^3$.

The source proves nearby low-dimensional and two-segment results. The theorem below gives the conjectured conclusion in every dimension for a natural regular-matroid class of finite zonotopes.

2 Definitions

Definition 1. A list of nonzero vectors $w_1, \dots, w_N \in \mathbb{R}^n$ is *projectively unimodular* if there are an invertible $T \in \text{GL}_n(\mathbb{R})$, positive scalars $c_1, \dots, c_N$, and the columns $a_1, \dots, a_N$ of a full-row-rank totally unimodular $n \times N$ matrix A such that

$$w_e = c_e T a_e \quad (1 \leq e \leq N).$$

Replacing a generator by its negative only translates the corresponding segment, so it leaves all relevant volumes unchanged; signs may therefore be absorbed into the columns of A . Reduced oriented incidence matrices of graphs are totally unimodular, hence graphical zonotopes supply a large family of examples.

3 Proof intuition

For a general zonotope, its volume is a sum of absolute maximal determinants. Absolute values normally prevent that sum from being a single determinant. Projective unimodularity removes precisely this obstruction: every maximal minor of A is 0 or ± 1 , so absolute minors equal squared minors. Cauchy–Binet then packages the entire volume polynomial into a positive-semidefinite determinant.

After the base generators receive weight 1 and the m distinguished generators receive weight t , the polynomial is $\det(B + tC)$ with $B > 0$ and $C \geq 0$. Such a polynomial has only negative real roots. The m -th root of a product of at most m positive affine factors is concave by Cauchy–Schwarz. This last observation also handles the case in which the degree is smaller than m .

4 Main theorem

Theorem 2. *Let*

$$Z = \sum_{i=1}^r [0, v_i] \subset \mathbb{R}^n$$

be full-dimensional, let

$$L = \sum_{j=1}^m [0, u_j], \quad 1 \leq m \leq n,$$

and suppose that the combined list $(v_1, \dots, v_r, u_1, \dots, u_m)$ is projectively unimodular. Then

$$P(t) = \text{Vol}_n(Z + tL)$$

has only negative real zeros. Consequently $P(t)^{1/m}$ is concave on $[0, \infty)$.

Proof. Write $N = r + m$, let $w_1, \dots, w_N$ denote the combined list, and choose $T, c_e, A = (a_1 \cdots a_N)$ as in the definition. The standard zonotope volume formula gives, for nonnegative weights x_e ,

$$\begin{aligned} \text{Vol}_n \left(\sum_{e=1}^N x_e [0, w_e] \right) &= \sum_{\substack{S \subseteq [N] \\ |S|=n}} |\det(w_e : e \in S)| \prod_{e \in S} x_e \\ &= |\det T| \sum_{\substack{S \subseteq [N] \\ |S|=n}} |\det A_S| \prod_{e \in S} c_e x_e. \end{aligned} \tag{1}$$

Total unimodularity says $\det A_S \in \{0, \pm 1\}$, and hence $|\det A_S| = (\det A_S)^2$. Cauchy–Binet turns (1) into

$$\text{Vol}_n \left(\sum_{e=1}^N x_e [0, w_e] \right) = |\det T| \det \left(A \text{diag}(c_e x_e) A^\top \right). \tag{2}$$

Partition $A = (A_Z \ A_L)$ and the c_e ’s according to the generators of Z and L . Setting the Z -weights to 1 and the L -weights to t in (2) yields

$$P(t) = |\det T| \det(B + tC),$$

where

$$B = A_Z \text{diag}(c_1, \dots, c_r) A_Z^\top, \quad C = A_L \text{diag}(c_{r+1}, \dots, c_{r+m}) A_L^\top.$$

Both matrices are positive semidefinite. Since Z is full-dimensional, the vectors $v_1, \dots, v_r$, and therefore the columns of A_Z , span $\mathbb{R}^n$. Thus B is positive definite.

If $\text{Im } t > 0$ and $(B + tC)q = 0$ for a complex vector q , then taking the imaginary part of $q^*(B + tC)q = 0$ gives

$$(\text{Im } t) q^* C q = 0.$$

Hence $Cq = 0$, and then $q^* B q = 0$, which forces $q = 0$. Therefore $\det(B + tC) \neq 0$ in the open upper half-plane. Because P has real coefficients, any nonreal zero would bring its conjugate into that half-plane. Thus all zeros of P are real. Its coefficients are nonnegative and $P(0) = \text{Vol}_n(Z) > 0$, so all its zeros are strictly negative.

Let $d = \deg P \leq m$. There are $\rho_1, \dots, \rho_d > 0$ and $K > 0$ such that

$$P(t) = K \prod_{\ell=1}^d (t + \rho_\ell).$$

For $F = P^{1/m}$ and $t \geq 0$,

$$\frac{F''(t)}{F(t)} = -\frac{1}{m} \sum_{\ell=1}^d \frac{1}{(t + \rho_\ell)^2} + \frac{1}{m^2} \left(\sum_{\ell=1}^d \frac{1}{t + \rho_\ell} \right)^2.$$

Cauchy–Schwarz gives

$$\left(\sum_{\ell=1}^d \frac{1}{t + \rho_\ell} \right)^2 \leq d \sum_{\ell=1}^d \frac{1}{(t + \rho_\ell)^2} \leq m \sum_{\ell=1}^d \frac{1}{(t + \rho_\ell)^2}.$$

It follows that $F''(t) \leq 0$, as claimed. $\square$

Corollary 3 (Graphical zonotopes). *Let G be a connected graph on $n + 1$ vertices and represent each edge by its column in a reduced oriented incidence matrix. Partition the edges as $E = E_Z \sqcup E_L$, where the subgraph (V, E_Z) is connected and $|E_L| = m \leq n$. Positive rescalings of the edge vectors are allowed. If Z is generated by E_Z and L by E_L , then $\text{Vol}_n(Z + tL)^{1/m}$ is concave for $t \geq 0$.*

Proof. A reduced oriented incidence matrix is totally unimodular. Connectivity of (V, E_Z) gives full rank to the Z -columns. Apply Theorem 2. $\square$

5 Exact four-dimensional example

Example 4. Take the reduced incidence matrix of K_5 , so the ambient dimension is 4. Let the three distinguished edges be

$$E_L = \{\{0, 1\}, \{0, 2\}, \{0, 3\}\}$$

and let the other seven edges generate Z . The latter subgraph is connected, and exact Cauchy–Binet evaluation gives

$$P(t) = 4t^3 + 33t^2 + 72t + 16 = (t + 4)^2(4t + 1).$$

This is a non-parallelotope example in the first dimension not covered by the source paper’s $\mathbb{R}^3$ result. Directly,

$$3P(t)P''(t) - 2P'(t)^2 = -450(t + 4)^2 \leq 0,$$

which is the differential inequality for concavity of $P^{1/3}$. The exact verifier included with this packet checks all of these identities over the integers.

6 Verification and limitations

Adversarial checks

Potential failure	Check
Missing determinant normalization	For segments $[0, w_e]$, the zonotope formula is exactly the sum of absolute maximal determinants; there is no factorial.
Absolute versus squared minors	The equality uses total unimodularity explicitly. It is false for a general vector configuration.
Singular base pencil	Full-dimensionality of Z is used to prove $B > 0$.
Degree smaller than m	The final Cauchy–Schwarz bound uses $d \leq m$, not $d = m$.
Complex-kernel argument	B, C are real symmetric positive semidefinite, so $q^*Cq = 0$ implies $Cq = 0$, and $q^*Bq = 0$ implies $q = 0$.

Scope

The theorem does not settle the source conjecture for arbitrary zonoids, or even arbitrary finite zonotopes. Its hypothesis concerns the combined directions of Z and L . A tempting attempt to divide a forced monomial from the general degree- n Lorentzian volume polynomial fails because it does not preserve the stronger degree- m Newton normalization.

Bounded novelty check

On 23 July 2026, searches were run for the exact source conjecture and arXiv id, and for combinations of *zonoid*, *totally unimodular*, *projectively unimodular*, *graphical zonotope*, *volume polynomial*, *real stable*, and *concavity*. The search recovered the source paper and general literature on Lorentzian and volume polynomials, including [2], but no statement of the precise subcase above. This is a bounded negative search, so novelty confidence is moderate rather than definitive.

Human-review recommendation

Send to a human reviewer. The key steps to inspect are (2), the use of full-dimensionality to obtain $B > 0$, and the exact scope of the projectively-unimodular hypothesis. If accepted, the result is suitable as a short proposition or note; the general conjecture should continue to be presented as open.

References

- [1] M. Fradelizi, M. Madiman, M. Meyer, and A. Zvavitch, *On the volume of the Minkowski sum of zonoids*, J. Funct. Anal. **286** (2024), 110247; arXiv:2206.02123.
- [2] P. Brändén and J. Huh, *Lorentzian polynomials*, Ann. of Math. **192** (2020), 821–891; arXiv:1902.03719.